

ASCENT-3 — Statistical Analysis Plan (Synopsis)

Protocol RZB441-UC-301 | Phase III, randomized, double-blind, placebo-controlled trial of RZB-441 (roزالibumab) in moderate-to-severe ulcerative colitis | SAP Version 3.0 | Status: FINAL, approved pre-database-lock

This SAP synopsis is the controlling methodological reference for the Week-24 interim integrity re-analysis. Where any working draft, summary table, or export disagrees with the methods defined here, this SAP and the subject-level source data govern.

1. Study Design Parameters

Parameter	Specification
Design	Randomized 1:1, double-blind, placebo-controlled, parallel-group
Treatment arms	RZB-441 (active) and Placebo — these are the ONLY two valid arm codes
Randomization stratification factors	Geographic region (3 levels) and prior biologic exposure (Naive / Experienced)
Planned randomized N	480 (240 per arm)
Study start (first subject randomized)	2024-01-15
Interim data cutoff (DSMB)	2025-09-30 — records randomized after this date are not part of the interim
Primary endpoint	Time to confirmed clinical remission by Week 24 (Day 168)
Key secondary endpoint	Clinical response at Week 8 (binary)
Pre-specified MCID (remission rate diff.)	10.0 percentage points

2. Analysis Populations and Record-Validity Rules

Intention-to-treat (ITT) — PRIMARY analysis population. All validly randomized subjects, analyzed as randomized. ITT is the primary population for all efficacy endpoints. The Per-Protocol (PP) set is supportive only and must NOT be substituted for ITT in the primary efficacy or go/no-go assessment (ICH E9(R1)).

Record-validity rules (applied before any population is defined). A record is invalid and must be excluded from ALL analysis populations if any of the following holds. The data-management query log (ascent3_data_query_log.csv) documents each affected subject:

#	Invalidity rule	Evidence source
1	Duplicate subject_id (retain first valid occurrence only)	subject_level
2	randomization_date after the 2025-09-30 interim cutoff	subject_level
3	arm not equal to 'RZB-441' or 'Placebo' (incl. blank / 'Screen Failure' / off-protocol dose)	subject_level
4	death_date earlier than randomization_date	subject_level
5	subject_id absent from the IWRS randomization schedule	randomization_schedule
6	any visit_date earlier than randomization_date	visit_records

Per-Protocol (PP) set. ITT minus subjects with a major protocol deviation flagged 'PP-EXCLUDE' in the query log (prohibited concomitant biologic, study-drug compliance < 80%, randomized but never dosed, or wrong stratum entered). Major deviations remain in ITT.

3. Primary Endpoint — Estimand and Estimation Method

Estimand. Cumulative incidence of confirmed clinical remission by Week 24 (Day 168) in the ITT population, by arm, with the treatment effect expressed as the between-arm remission-rate difference (RZB-441 minus Placebo).

Estimation method — Kaplan-Meier (mandatory). Because subjects discontinue before Week 24 and follow-up is incomplete, the Week-24 remission rate **MUST** be estimated using the Kaplan-Meier estimator [$F(168) = 1 - S(168)$], censoring subjects at their last on-study assessment (last_contact_day). The naive crude proportion (confirmed remissions ÷ randomized) and the completers-only proportion are **BIASED** in the presence of differential dropout and are **NOT** acceptable as the primary estimate (see Kaplan-Meier-for-proportions reference, External Links). Report each arm's KM rate with a two-sided 95% confidence interval using Greenwood's variance, and the rate difference with its 95% CI.

4. Primary Success (Go / No-Go) Decision Rule

The interim outcome is declared **GO** only if **BOTH** conditions are met, evaluated on the Kaplan-Meier ITT estimate of the Week-24 remission-rate difference:

Condition	Requirement
(i) Statistical superiority	Lower bound of the two-sided 95% CI for the KM rate difference is > 0
(ii) Clinical relevance	Point estimate of the KM rate difference is ≥ 10.0 percentage points (MCID)

If condition (i) holds but condition (ii) does not, the result is a statistically significant but not clinically meaningful effect and the decision is **NO-GO**. A crude or completers-based rate difference must not be used to evaluate this rule.

5. Secondary Endpoint and Severity-Adjusted Analysis

Week-8 clinical response (binary) is analyzed in the ITT population. Baseline disease severity (Moderate vs Severe, from baseline total Mayo score) is a strong prognostic factor that was **NOT** a randomization stratification factor and may be imbalanced between arms. The SAP therefore pre-specifies a **severity-adjusted analysis using the Cochran-Mantel-Haenszel (CMH) common risk difference**, stratified by baseline severity, as the governing estimate for the Week-8 endpoint. Report the crude pooled rate difference, the within-stratum rate differences, and the CMH-adjusted rate difference. **Where the crude pooled estimate and the severity-adjusted estimate diverge by more than 5 percentage points (Simpson's-paradox reversal), the CMH-adjusted estimate governs the interpretation** and the crude pooled estimate must be flagged as confounded.

6. Multi-Center Pooling Rule

Any rate pooled across sites or regions must be a **sample-size-weighted** estimate, i.e. the pooled proportion equals total responders ÷ total subjects across the pooled units. The unweighted arithmetic mean of per-site rates is **NOT** a valid pooled estimate and must not be reported as the overall rate, because site sample sizes differ materially.

7. Sample-Size Methodology for the Confirmatory Study

The planned 52-week confirmatory study is sized for the two-proportion (remission rate) comparison using:

$$n \text{ per arm} = (Z_{1-\alpha/2} + Z_{1-\beta})^2 \times [p_1(1-p_1) + p_2(1-p_2)] \div (p_1 - p_2)^2$$

Assumption	Value
Two-sided alpha	0.05 (Z = 1.9600)
Power	0.9 (Z = 1.2816)
Dropout inflation	20% (divide computed n by 0.80)
Originally assumed remission rates	p1 = 0.40 (RZB-441), p2 = 0.25 (Placebo)
Re-estimation requirement	Re-size using the corrected Week-24 KM remission rates from this interim

Round each computed per-arm n UP to the next whole subject before and after dropout inflation.

8. Regulatory References

ICH E9(R1) Addendum on Estimands and Sensitivity Analysis (analysis populations, ITT principle, estimands). CONSORT 2010 Statement (subject disposition / flow accounting and reporting of exclusions and baseline imbalance). See the task External Links for URLs.